

The Sporadic Apéry Atlas

Hecke–Eisenstein Sources, Periods, and the Zagier Recurrence Census

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Abstract

The classical “sporadic Apéry-like sequences” form one of the most striking finite collections in experimental arithmetic: six second-order sequences isolated by Zagier, six closely related cubic sequences of Almkvist–Zudilin type, and three further cubic systems of Cooper. We give a uniform structural atlas of this standard fifteen-family census using the Hecke–Eisenstein–Apéry framework developed in [1]. For the twelve non-Cooper modular rows, the reverse machine is exact: if $t(q)$ is the canonical modular coordinate, $F(q)$ the principal period, and $D = q d/dq$, then the canonical extension source is

$$\Phi = F D t,$$

and is a pure holomorphic Eisenstein form. We record the exact eta-quotient parametrizations, levels and characters, divisor-sum formulas for every source, their Dirichlet-series factorizations, and the resulting critical periods. The normalized companion is universally

$$y_B(t(q)) = F(q) D^{-r} \Phi(q),$$

with $r = 2$ or 3 , so the second solution is the same modular period multiplied by an Eisenstein Eichler integral.

The six second-order rows and six cubic rows pair under the exact symmetric-square map

$$(a, b, c) = (A, A - 2\lambda, A^2 - 4B),$$

reducing twelve recurrences to six underlying rank-two geometries. At Layer B the collapse is not trivial: among the four untwisted cubic $\zeta(3)$ sources, three are full period projectors, while four normalized extension classes form an affine three-dimensional family over a single homogeneous modular-correspondence class. The three Cooper rows belong to a different generalized symmetric-square stratum. Their first nominal Eichler integration is arithmetically free, giving the sharp denominator weight 2 despite differential order 3.

We also classify Zagier’s two infinite hypergeometric/Legendrian families. They are the same Gauss local system under a Pfaff pullback and gauge: the hypergeometric companion is rationally degenerate, while the Legendrian companion has an exact trigamma/cyclotomic L -value limit but only polynomial separation. This explains, within the three-layer framework, why the isolated modular rows are the natural home of the classical Apéry phenomenon. The paper classifies the standard known fifteen-family census and the two infinite families; it does not claim to prove Zagier’s stronger global enumeration conjecture that no further integral recurrences exist.

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1 Introduction

Apéry-like recurrences sit at an unusually fertile intersection of combinatorics, modular forms, periods and Diophantine approximation. Their most visible feature is an integral solution A_n , but the arithmetic content is often carried by a second solution B_n and the connection constant measured by B_n/A_n . Zagier’s search [4] isolated a small number of exceptional second-order recurrences, Almkvist and Zudilin found cubic analogues in the mirror-symmetry setting [6], and Cooper added three

further modular cubic systems [7]. Modern literature normally refers to the resulting collection of fifteen as the sporadic Apéry-like sequences [11].

The usual presentation starts with a recurrence and then asks why it is modular. The point of the Hecke–Eisenstein–Apéry theory [1] is to reverse that logic. The fundamental object is not the recurrence itself but a three-layer realization

pure modular local system $\longrightarrow$ Eisenstein/polylogarithmic extension $\longrightarrow$ Apéry coordinate and integral realization

The recurrence is a shadow of this geometry in a particular coordinate. In the present paper we apply this philosophy exhaustively to the classical sporadic census.

The result is a reference atlas rather than a list of coincidences. Each row is assigned: its recurrence, canonical modular coordinate t , principal period F , source $\Phi = F D t$, Eisenstein sector, critical L -value, extension depth, denominator weight, fold geometry, and known congruence properties. The striking compression is that twelve of the fifteen rows are controlled by six rank-two geometries, while the remaining three form a coherent Cooper branch.

1.1 What is and is not being classified

There are two logically different completeness assertions in the subject.

First, there is a finite *standard census*: the six Zagier rows $A, \dots, F$, six Almkvist–Zudilin rows $\alpha, \gamma, \delta, \epsilon, \zeta, \eta$, and Cooper’s s_7, s_{10}, s_{18} . This paper gives a complete structural classification of that fifteen-family set, together with Zagier’s two infinite hypergeometric/Legendrian families.

Second, Zagier’s original work raises a stronger global enumeration problem: whether the search families and isolated examples exhaust every integral recurrence in the relevant normalization. That statement is not proved here, and we will not use the word “classification” to disguise this distinction. The final section formulates the converse enumeration problem in the language of the three-layer theory.

1.2 Main structural theorem

Theorem 1.1 (Sporadic atlas theorem). *Let $\mathcal{S}_{15}$ be the standard fifteen sporadic Apéry-like systems. Then:*

(i) *For the twelve rows*

$$A, B, C, D, E, F, \alpha, \gamma, \delta, \epsilon, \zeta, \eta,$$

the canonical reverse-machine source satisfies

$$\Phi = F D t,$$

and is a holomorphic pure Eisenstein modular form of weight $r + 1$, where $r = 2$ for $A, \dots, F$ and $r = 3$ for the six cubic rows. Its coefficients and critical period are given explicitly in Tables 3 and 4.

(ii) *The normalized companion with $B_0 = 0, B_1 = 1$ is*

$$y_B(t(q)) = F(q) D^{-r} \Phi(q).$$

Consequently $d_n^r B_n \in \mathbb{Z}$ for $d_n = \text{lcm}(1, \dots, n)$ whenever the integral mirror map is used.

(iii) *The six second-order and six classical cubic rows are paired by*

$$(a, b, c) = (A, A - 2\lambda, A^2 - 4B),$$

with

$$A \leftrightarrow \delta, \quad B \leftrightarrow \zeta, \quad C \leftrightarrow \alpha, \quad D \leftrightarrow \eta, \quad E \leftrightarrow \epsilon, \quad F \leftrightarrow \gamma.$$

The cubic operator is the symmetric square of the associated rank-two operator. Thus twelve recurrence rows reduce to six homogeneous rank-two source geometries.

- (iv) The three Cooper rows are exact generalized symmetric squares and still satisfy $\Phi = FDt$, but their natural source need not lie in the same holomorphic Eisenstein stratum. They possess an integral first primitive $D^{-1}\Phi$; equivalently, $(n+1) \mid A_n$. Hence

$$d_n^2 B_n \in \mathbb{Z}$$

for all three, and the exponent 2 is sharp.

- (v) The two infinite families HG_d and Leg_d are the same Gauss hypergeometric local system under a Möbius pullback and rank-one gauge. The normalized HG_d companion is rationally proportional to the first solution, whereas the Leg_d companion has the exact limit

$$\xi_d = 2 \sum_{j \geq 0} \frac{(-1)^j}{(j+d+1)^2} = \frac{1}{2} \left[\psi_1\left(\frac{d+1}{2}\right) - \psi_1\left(\frac{d+2}{2}\right) \right],$$

but the corrected solution has only polynomial, not exponential, separation.

Parts (i)–(iii) are the finite-source application of [1]; part (iv) is the free-integration refinement of the generalized symmetric-square mechanism; part (v) is proved in Section 11. The point of collecting them in one theorem is that they answer a single structural question: what distinguishes the isolated sporadic rows from the generic recurrence families?

2 Recurrences and normalizations

2.1 Second-order normalization

We use Zagier's convention

$$(n+1)^2 A_{n+1} = (An^2 + An + \lambda)A_n - Bn^2 A_{n-1}, \quad A_0 = 1. \quad (1)$$

The six standard rows are

row	A	B	λ
A	7	-8	2
B	9	27	3
C	10	9	3
D	11	-1	3
E	12	32	4
F	17	72	6

The corresponding differential operator in $\theta = t d/dt$ is

$$L_2 = \theta^2 - t(A\theta^2 + A\theta + \lambda) + Bt^2(\theta + 1)^2.$$

2.2 Cubic and Cooper normalization

For all nine cubic rows we use the single sign convention

$$(n+1)^3 A_{n+1} = (2n+1)(an^2 + an + b)A_n - n(cn^2 + d)A_{n-1}. \quad (2)$$

Thus the six classical rows have $d = 0$, while the Cooper cases allow $d \neq 0$. The data are

row	a	b	c	d
δ	7	3	81	0
ζ	9	3	-27	0
α	10	4	64	0
η	11	5	125	0
ϵ	12	4	16	0
γ	17	5	1	0
s_7	13	4	-27	3
s_{10}	6	2	-64	4
s_{18}	14	6	192	-12

Cooper's own recurrence is often written with a plus sign in front of the final term; under that convention the triples are $(13, 4, 27, -3)$, $(6, 2, 64, -4)$ and $(14, 6, -192, 12)$ respectively.

The differential operator attached to (2) is

$$L_3 = \theta^3 - t(2\theta + 1)(a\theta^2 + a\theta + b) + t^2(\theta + 1)(c(\theta + 1)^2 + d). \quad (3)$$

Proposition 2.1 (Exact symmetric square). *The operator (3) is the symmetric square of*

$$\tilde{L}_2 = \theta^2 - t\left(2a\theta^2 + a\theta + \frac{b}{2}\right) + t^2\left(c\left(\theta + \frac{1}{2}\right)^2 + \frac{d}{4}\right).$$

For $d = 0$ this gives the six classical cubic rows; for $d \neq 0$ it gives all three Cooper rows.

The identity is algebraic in a, b, c, d and can be checked by expanding the symmetric square formula for a second-order differential equation. Its importance is conceptual: the cubic rows are not new homogeneous geometries of rank three; they are symmetric-square realizations of rank-two data.

2.3 The six pairings

If (A, B, λ) is a second-order row, then

$$(a, b, c) = (A, A - 2\lambda, A^2 - 4B) \quad (4)$$

produces its classical cubic partner. Direct substitution gives

$$A \leftrightarrow \delta, \quad B \leftrightarrow \zeta, \quad C \leftrightarrow \alpha, \quad D \leftrightarrow \eta, \quad E \leftrightarrow \epsilon, \quad F \leftrightarrow \gamma.$$

This relation, already visible in the literature, acquires a sharper meaning in the three-layer classification: the pair shares its underlying rank-two homogeneous system, while the extension and Apéry realization may differ.

3 The reverse machine

Let $y_0(t) = 1 + O(t)$ be the holomorphic first solution and let

$$y_1(t) = y_0(t) \log t + g(t)$$

be the logarithmic Frobenius solution. The canonical nome is

$$q = t \exp(g/y_0), \quad t = t(q), \quad F(q) = y_0(t(q)),$$

and we put $D = q d/dq$ and

$$J = \frac{Dt}{t}.$$

For an order- r Apéry operator whose leading θ^r coefficient is $P(t)$, exact rectification in the canonical coordinate has the form

$$L = C(t)FD^rF^{-1}, \quad C(t) = \frac{P(t)}{J^r}.$$

The normalized companion is the unique log-free solution of

$$Ly_B = t, \quad y_B(0) = 0.$$

Variation of constants gives the source

$$\Phi = \frac{tJ^r}{P(t)F}, \quad y_B(t(q)) = F(q)D^{-r}\Phi(q). \quad (5)$$

Theorem 3.1 (Source collapse). *For each of the twelve classical modular rows, and for the generalized symmetric-square Cooper rows, the Wronskian identity reduces (5) to*

$$\boxed{\Phi = F Dt.}$$

Proof. For a second-order row the Wronskian gives

$$Dt = tP(t)F^2.$$

Thus $J = P(t)F^2$ and

$$\frac{tJ^2}{P(t)F} = tP(t)F^3 = FDt.$$

For a symmetric-square cubic row, write $F = u_0^2$ for the square of the normalized rank-two period. The rank-two Wronskian yields

$$(Dt)^2 = t^2P(t)F^2,$$

so $J^2 = P(t)F^2$ and

$$\frac{tJ^3}{P(t)F} = tJF = FDt.$$

The extra Cooper d -term changes the potential term of the rank-two equation but not the Wronskian coefficient; hence the same calculation applies. $\square$

The formula is the basic bridge between recurrence and modular source. Once t and F are known, no separate search for the inhomogeneous extension is necessary.

4 The master census

Table 1: Structural census of the standard fifteen sporadic systems.
“Source period” means the canonical Mellin/Eichler value of Φ ; it is
logically distinct from a coefficient-ratio limit when dominant folds are
non-real.

row	ord.	level	source stratum	source period / connection content	dominant-fold behavior	comp. wt.
A	2	6	holomorphic Eisenstein	$\zeta(2)/4$	real; ordinary Apéry limit	2
B	2	36	holomorphic Eisenstein	$\frac{1}{2}L(2, \chi_{-3})$	complex-conjugate dominant folds; no single real ratio limit	2
C	2	6	holomorphic Eisenstein	$\frac{1}{2}L(2, \chi_{-3})$	real; ordinary Apéry limit	2
D	2	5	$\Gamma_1(5)$ Eisenstein	$\zeta(2)/5$	real; classical Apéry $\zeta(2)$ system	2
E	2	8	holomorphic Eisenstein	$\frac{1}{2}L(2, \chi_{-4}) = G/2$	real; Catalan connection	2
F	2	12	holomorphic Eisenstein	$\frac{5}{8}L(2, \chi_{-3})$	real	2
α	3	12	old- E_4 Eisenstein	$\frac{7}{24}\zeta(3)$	real	3
γ	3	6	old- E_4 Eisenstein	$\frac{1}{6}\zeta(3)$	real; Apéry $\zeta(3)$ after scaling	3
δ	3	12	old- E_4 Eisenstein	$\frac{13}{54}\zeta(3)$	complex dominant pair in canonical realization	3
ϵ	3	8	old- E_4 Eisenstein	$\frac{7}{32}\zeta(3)$	real	3
ζ	3	9	$\chi_{-3} \otimes \chi_{-3}$ Eisenstein	$\frac{1}{3}L(3, \chi_{-3}) = \frac{4\pi^3}{243\sqrt{3}}$	real	3
η	3	20	χ_5 -twisted Eisenstein	$\frac{1}{2}L(3, \chi_5)$	complex dominant pair in canonical realization	3
s_7	3	7	Cooper generalized symmetric square	numerically $\zeta(2)/7$ connection; Fricke fixed dominant fold	real; exact published modular parametrization	2 sharp
s_{10}	3	10	Cooper generalized symmetric square	$\zeta(2)/5$ (published Apéry limit)	real	2 sharp
s_{18}	3	18	Cooper generalized symmetric square	numerically $\frac{1}{2}L(2, \chi_{-3})$ connection	real, slower convergence	2 sharp

The master table deliberately separates exact source values from archimedean connection claims. For the twelve holomorphic-source rows, the source period is an exact theorem. For B, δ, η , a complex-conjugate dominant pair prevents a single ordinary real coefficient ratio from converging even though the Eichler source is completely explicit. For s_7 and s_{18} , the displayed constants are strong numerical connection values; we do not promote them to theorems without a complete Fricke/Eichler evaluation.

5 Exact modular parametrizations

Write $\eta_d = \eta(d\tau)$. The following table records the canonical coordinates for the eleven eta-quotient rows. These identities are certified by eta-quotient modularity, cusp-order analysis, the differential equation/Wronskian identity, and a finite Sturm comparison; see Appendix A.

Table 2: Canonical eta-quotient coordinates and principal periods.

row	level	$t(\tau)$	$F(\tau)$
A	6	$\eta_1^3 \eta_6^9 / (\eta_2^3 \eta_3^9)$	$\eta_2 \eta_3^6 / (\eta_1^2 \eta_6^3)$
B	36	$\eta_1^3 \eta_4^3 \eta_{18}^9 / (\eta_2^9 \eta_9^3 \eta_{36}^3)$	$\eta_2^9 \eta_3 \eta_{12} / (\eta_1^3 \eta_4^3 \eta_6^3)$
C	6	$\eta_1^4 \eta_6^8 / (\eta_2^8 \eta_3^4)$	$\eta_2^6 \eta_3 / (\eta_1^3 \eta_6^6)$
E	8	$\eta_1^4 \eta_4^4 \eta_8^4 / \eta_2^{10}$	$\eta_2^{10} / (\eta_1^4 \eta_4^4)$
F	12	$\eta_1^3 \eta_3 \eta_4^3 \eta_6^2 \eta_{12} / \eta_2^{14}$	$\eta_2^{15} \eta_3^2 \eta_{12}^2 / (\eta_1^6 \eta_4^6 \eta_6^5)$
α	12	$\eta_1^6 \eta_3^6 \eta_4^6 \eta_{12}^6 / (\eta_2^{12} \eta_6^{12})$	$\eta_2^{10} \eta_6^{10} / (\eta_1^4 \eta_3^4 \eta_4^4 \eta_{12}^4)$
γ	6	$(\eta_1 \eta_6 / (\eta_2 \eta_3))^{12}$	$(\eta_2 \eta_3)^7 / (\eta_1 \eta_6)^5$
δ	12	$\eta_1^4 \eta_4^4 \eta_6^{16} / (\eta_2^{16} \eta_3^4 \eta_{12}^4)$	$\eta_2^{12} \eta_3 \eta_{12} / (\eta_1^3 \eta_4^3 \eta_6^4)$
ϵ	8	$\eta_1^8 \eta_8^8 / (\eta_2^8 \eta_4^8)$	$\eta_2^6 \eta_4^6 / (\eta_1^4 \eta_8^4)$
ζ	9	$\eta_1^6 \eta_9^6 / \eta_3^{12}$	$\eta_3^{10} / (\eta_1^3 \eta_9^3)$
η	20	$\eta_1^6 \eta_4^6 \eta_{10}^{18} / (\eta_2^{18} \eta_5^6 \eta_{20}^6)$	$\eta_2^{15} \eta_5 \eta_{20} / (\eta_1^5 \eta_4^5 \eta_{10}^3)$

The missing D row is genuinely different rather than an omission. Zagier's $\Gamma_1(5)$ parametrization may be normalized as

$$t = q \prod_{n \geq 1} (1 - q^n)^{5(\frac{n}{5})}, \quad Dt = t(1 - 11t - t^2)F^2, \quad (6)$$

where $(\frac{n}{5})$ is the quadratic symbol. The source FDt is a weight-three Eisenstein form on $\Gamma_1(5)$ built from the primitive quartic character modulo 5.

6 The twelve exact Eisenstein sources

For a Dirichlet character χ define

$$\sigma_{\text{in}}^{\chi,k}(m) = \sum_{d|m} \chi(m/d)d^k, \quad \sigma_{\text{out}}^{\chi,k}(m) = \sum_{d|m} \chi(d)d^k,$$

and set all shifted expressions at nonintegral arguments equal to zero. Also $\sigma_3(m) = \sum_{d|m} d^3$.

6.1 Second-order rows

Table 3: Exact weight-three source coefficients for the second-order rows.

row	$\Phi = \sum_{m \geq 1} c(m)q^m$	canonical critical value
A	$c = \sigma_{\text{out}}^{\chi_{-3},2}(m) - \sigma_{\text{out}}^{\chi_{-3},2}(m/2)$	$\zeta(2)/4$
B	$c = \sigma_{\text{in}}^{\chi_{-3},2}(m) - 6\sigma_{\text{in}}^{\chi_{-3},2}(m/2) - 8\sigma_{\text{in}}^{\chi_{-3},2}(m/4)$	$\frac{1}{2}L(2, \chi_{-3})$
C	$c = \sigma_{\text{in}}^{\chi_{-3},2}(m) - 8\sigma_{\text{in}}^{\chi_{-3},2}(m/2)$	$\frac{1}{2}L(2, \chi_{-3})$
D	$c = \sum_{d m} (\psi_1(d) - 2\psi_2(d))d^2$	$\zeta(2)/5$
E	$c = \sigma_{\text{in}}^{\chi_{-4},2}(m) - 8\sigma_{\text{in}}^{\chi_{-4},2}(m/2)$	$\frac{1}{2}L(2, \chi_{-4}) = G/2$
F	$c = \sigma_{\text{in}}^{\chi_{-3},2}(m) - 7\sigma_{\text{in}}^{\chi_{-3},2}(m/2) - 8\sigma_{\text{in}}^{\chi_{-3},2}(m/4)$	$\frac{5}{8}L(2, \chi_{-3})$

For D , choose the primitive quartic character ρ modulo 5 with $\rho(2) = i$, and write $\psi_1 = \Re\rho$, $\psi_2 = \Im\rho$.

6.2 Cubic rows

Table 4: Exact weight-four source coefficients for the six classical cubic rows.

row	$c(m)$	canonical critical value
α	$\sigma_3(m) - 17\sigma_3(m/2) - 9\sigma_3(m/3) + 16\sigma_3(m/4) + 153\sigma_3(m/6) - 144\sigma_3(m/12)$	$\frac{7}{24}\zeta(3)$
γ	$\sigma_3(m) - 28\sigma_3(m/2) + 63\sigma_3(m/3) - 36\sigma_3(m/6)$	$\frac{1}{6}\zeta(3)$
δ	$\sigma_3(m) - 14\sigma_3(m/2) - \sigma_3(m/3) + 16\sigma_3(m/4) + 14\sigma_3(m/6) - 16\sigma_3(m/12)$	$\frac{13}{54}\zeta(3)$
ϵ	$\sigma_3(m) - 21\sigma_3(m/2) + 84\sigma_3(m/4) - 64\sigma_3(m/8)$	$\frac{7}{32}\zeta(3)$
ζ	$\sum_{d m} \chi_{-3}(d)\chi_{-3}(m/d)d^3$	$\frac{1}{3}L(3, \chi_{-3}) = \frac{4\pi^3}{243\sqrt{3}}$
η	$\sigma_{\text{in}}^{\chi_5,3}(m) - 14\sigma_{\text{in}}^{\chi_5,3}(m/2) - 16\sigma_{\text{in}}^{\chi_5,3}(m/4)$	$\frac{1}{2}L(3, \chi_5)$

6.3 Why the periods are visible immediately

The divisor sums factor multiplicatively:

$$\begin{aligned}\sum_{n \geq 1} \frac{\sigma_3(n)}{n^s} &= \zeta(s)\zeta(s-3), \\ \sum_{n \geq 1} \frac{\sigma_{\text{in}}^{\chi,k}(n)}{n^s} &= L(s, \chi)\zeta(s-k), \\ \sum_{n \geq 1} \frac{\sigma_{\text{out}}^{\chi,k}(n)}{n^s} &= \zeta(s)L(s-k, \chi).\end{aligned}$$

An oldform shift $a(n/d)$ contributes the scalar d^{-s} . Thus if

$$\Phi = \sum_d c_d V_d E,$$

then

$$L(\Phi, s) = P_\Phi(s)L(E, s), \quad P_\Phi(s) = \sum_d c_d d^{-s}.$$

This is exactly the finite Hecke/Eisenstein period projector of [1]. The special constant is therefore not recognized numerically after the recurrence has been found: it is already encoded by the finite oldform polynomial.

7 A worked row: Apéry's $\zeta(3)$ recurrence

The γ row is the most transparent example. Its canonical data are

$$t = \left(\frac{\eta(\tau)\eta(6\tau)}{\eta(2\tau)\eta(3\tau)} \right)^{12}, \quad F = \frac{(\eta(2\tau)\eta(3\tau))^7}{(\eta(\tau)\eta(6\tau))^5}.$$

The exact source identity is

$$\Phi_\gamma = F D t = \frac{E_4(\tau) - 28E_4(2\tau) + 63E_4(3\tau) - 36E_4(6\tau)}{240}. \quad (7)$$

Hence

$$L(\Phi_\gamma, s) = (1 - 28 \cdot 2^{-s} + 63 \cdot 3^{-s} - 36 \cdot 6^{-s}) \zeta(s)\zeta(s-3).$$

The finite multiplier has

$$P(0) = P(2) = P(4) = 0, \quad P(3) = -\frac{1}{3},$$

so

$$L(\Phi_\gamma, 3) = \frac{1}{6}\zeta(3).$$

The normalized companion is therefore

$$y_B(t(q)) = F(q)\Theta_\gamma(q),$$

with

$$\Theta_\gamma = \mathcal{L}_3(q) - \frac{7}{2}\mathcal{L}_3(q^2) + \frac{7}{3}\mathcal{L}_3(q^3) - \frac{1}{6}\mathcal{L}_3(q^6), \quad (8)$$

where

$$\mathcal{L}_3(q) = \sum_{m \geq 1} \frac{\sigma_3(m)}{m^3} q^m = \sum_{d \geq 1} \text{Li}_3(q^d).$$

Thus Apéry's second solution is literally the modular period multiplied by a four-term trilogarithmic Eichler integral. In the historical normalization $b_1 = 6$, one has $b_n = 6B_n$ and the coefficient ratio tends to $\zeta(3)$ rather than $\zeta(3)/6$.

This row contains the entire reverse-machine philosophy in miniature:

$$\zeta(3) \rightsquigarrow P(s)\zeta(s)\zeta(s-3) \rightsquigarrow \Phi_\gamma \rightsquigarrow (t, F) \rightsquigarrow \text{Apéry recurrence}.$$

8 The six paired geometries

The map (4) means that the twelve classical rows should not be regarded as twelve unrelated modular miracles. They are six pairs obtained by changing the representation of a rank-two elliptic system.

Table 5: Pairing of the second-order and classical cubic rows.

rank-two row	cubic row	second-order level	cubic level
A	δ	6	12
B	ζ	36	9
C	α	6	12
D	η	5	20
E	ϵ	8	8
F	γ	12	6

The levels need not match because the same local system may be presented after different pullbacks, isogenies, Atkin–Lehner quotients, or rank-one twists. The correct invariant is therefore not the displayed level but the homogeneous modular-correspondence class of Layer A. This is precisely why a classification by recurrence coefficients alone appears sporadic while a classification by local systems becomes much smaller.

9 The cubic extension geometry: one homogeneous class, several extensions

The four untwisted cubic rows

$$\alpha, \gamma, \delta, \epsilon$$

all have sources in the rational old- E_4 space. Put the common divisor basis

$$d = (1, 2, 3, 4, 6, 8, 12, 24)$$

and write

$$\Phi_f = \frac{1}{240} \sum_d c_{f,d} E_4(d\tau).$$

The coefficient vectors are

$$c_\alpha = (1, -17, -9, 16, 153, 0, -144, 0),$$

$$c_\gamma = (1, -28, 63, 0, -36, 0, 0, 0),$$

$$c_\delta = (1, -14, -1, 16, 14, 0, -16, 0),$$

$$c_\epsilon = (1, -21, 0, 84, 0, -64, 0, 0).$$

For $P_f(s) = \sum_d c_{f,d} d^{-s}$ one finds

	$P(0)$	$P(2)$	$P(3)$	$P(4)$
α	0	0	$-7/12$	0
γ	0	0	$-1/3$	0
δ	0	$-4/3$	$-13/27$	$5/27$
ϵ	0	0	$-7/16$	0

Proposition 9.1 (Period-projector split). *Exactly α, γ, ϵ are full untwisted weight-four period projectors in the sense*

$$P(0) = P(2) = P(4) = 0.$$

The row δ is the unique untwisted $\zeta(3)$ source in this four-row set retaining nonzero unwanted critical directions.

The homogeneous geometry is nevertheless even more compressed.

Theorem 9.2 (Affine extension tetrahedron). *After pullback to common modular covers and the natural rank-one Kummer twists, the four untwisted cubic systems lie in one homogeneous Layer-A modular-correspondence class. After normalizing the target periods*

$$\lambda_\alpha = \frac{7}{24}, \quad \lambda_\gamma = \frac{1}{6}, \quad \lambda_\delta = \frac{13}{54}, \quad \lambda_\epsilon = \frac{7}{32},$$

their Layer-B Eichler extension classes are affinely independent. In particular their affine span has dimension 3.

Proof sketch with exact certificate. Use α as affine origin and form the integral relative sources

$$\Delta_{\alpha\gamma} = 4\Phi_\alpha - 7\Phi_\gamma, \quad \Delta_{\alpha\delta} = 52\Phi_\alpha - 63\Phi_\delta, \quad \Delta_{\alpha\epsilon} = 3\Phi_\alpha - 4\Phi_\epsilon.$$

In the divisor basis above their coefficient vectors are

$$\begin{aligned} &(-3, 128, -477, 64, 864, 0, -576, 0), \\ &(-11, -2, -405, -176, 7074, 0, -6480, 0), \\ &(-1, 33, -27, -288, 459, 256, -432, 0). \end{aligned}$$

The 3×3 minor on $d = (1, 2, 6)$ has determinant $128520 \neq 0$, so the sources are linearly independent. It remains to show that a nonzero rational old-Eisenstein source cannot disappear as an algebraic/Tate coboundary after D^{-3} . Choose a minimal divisor d_0 with coefficient $c_{d_0} \neq 0$. For all sufficiently large primes $p \nmid 24d_0$, the coefficient of $q^{d_0 p}$ in $D^{-3}\Phi$ contains a term proportional to $1 + p^{-3}$, producing infinitely many new denominator primes of valuation -3 . Algebraic modular sections have only finitely many denominator primes after a fixed normalization, while Tate polynomials have no positive Fourier tail. Hence no nonzero linear combination is a coboundary. The homogeneous equivalence follows from the common elliptic/isogeny presentations described in [1]; symmetric square transfers those rank-two equivalences to the cubic systems. $\square$

This is an instructive distinction between Layers A and B: one homogeneous geometry can support several genuinely distinct normal functions and therefore several different Apéry extensions.

10 The three Cooper systems

The Cooper rows are not exceptions to the reverse machine; they are exceptions to the *holomorphic source stratum*. Their coordinates are rational functions of Hauptmoduls, and FDt can acquire interior poles on the compactified modular curve. The correct classification therefore keeps the exact source identity but does not force these rows into the twelve-row pure Eisenstein theorem.

10.1 Published modular parametrizations

For s_7 , Cooper gives

$$\begin{aligned} x_7 &= \frac{\eta(\tau)^4}{\eta(7\tau)^4}, & t_7 &= \frac{x_7}{x_7^2 + 13x_7 + 49}, \\ F_7 &= \frac{1}{6}(7E_2(7\tau) - E_2(\tau)) = \sum_{n \geq 0} s_7(n)t_7^n, \end{aligned}$$

and

$$D \log t_7 = F_7 \sqrt{1 - 26t_7 - 27t_7^2} = F_7 \sqrt{(1 - 27t_7)(1 + t_7)}.$$

At the Fricke fixed point $\tau = i/\sqrt{7}$, one has $x_7 = 7$ and hence

$$t_7 = \frac{1}{27},$$

which is exactly the dominant singularity. This gives a geometric reason for the very stable numerical connection value $\zeta(2)/7$, although a complete proof of that connection formula requires the regularized Eichler transformation at the pole divisor.

For s_{10} ,

$$x_{10} = \frac{\eta(2\tau)^6 \eta(5\tau)^6}{\eta(\tau)^6 \eta(10\tau)^6}, \quad t_{10} = \frac{x_{10}}{(x_{10} - 1)^2},$$

$$F_{10} = \frac{1}{12} (10E_2(10\tau) + 5E_2(5\tau) - 2E_2(2\tau) - E_2(\tau)).$$

This is the fourth Franel sequence

$$s_{10}(n) = \sum_k \binom{n}{k}^4,$$

and its Apéry limit $\zeta(2)/5$ is covered by the general results of Straub–Zudilin [10].

For s_{18} ,

$$x_{18} = \frac{\eta(3\tau)^4 \eta(6\tau)^4}{\eta(\tau)^2 \eta(2\tau)^2 \eta(9\tau)^2 \eta(18\tau)^2}, \quad t_{18} = \frac{x_{18}}{(x_{18} + 3)^2},$$

$$F_{18} = \frac{1}{4} (18E_2(18\tau) - 9E_2(9\tau) - 12E_2(6\tau) + 6E_2(3\tau) + 2E_2(2\tau) - E_2(\tau)),$$

with

$$D \log t_{18} = F_{18} \sqrt{1 - 28t_{18} + 192t_{18}^2} = F_{18} \sqrt{(1 - 16t_{18})(1 - 12t_{18})}.$$

Numerically the normalized companion tends to $\frac{1}{2}L(2, \chi_{-3})$; we record this as a connection conjecture, not a proved theorem.

10.2 The free integration theorem

Theorem 10.1 (Cooper weight drop). *For s_7, s_{10}, s_{18} let $F(t) = \sum_{n \geq 0} A_n t^n$ and $\Phi = F D t$. Then*

$$D^{-1}\Phi = \int_0^t F(s) ds = \sum_{n \geq 0} \frac{A_n}{n+1} t^{n+1}.$$

For all three sequences $(n+1) \mid A_n$. Hence the first primitive is integral and the normalized cubic companion may be written

$$y_B = F D^{-2}(D^{-1}\Phi),$$

which gives

$$d_n^2 B_n \in \mathbb{Z}.$$

The exponent 2 is optimal.

Proof. The identity $d(D^{-1}\Phi)/dt = F$ follows from $D^{-1}(F D t)$ and the chain rule. The divisibility $(n+1) \mid A_n$ is the integral primitive property of the three Cooper sequences. Once the first primitive is integral, only two denominator-producing applications of D^{-1} remain. Since the mirror maps are integral, extracting the coefficient of t^n costs at most d_n^2 . Sharpness is already visible at $n = 2$:

$$B_2(s_7) = \frac{45}{4}, \quad B_2(s_{10}) = \frac{21}{4}, \quad B_2(s_{18}) = \frac{51}{4}.$$

Thus exponent 1 is impossible. □

This is a useful warning for any general Apéry theory: differential order and arithmetic denominator weight are not the same invariant. The true arithmetic invariant is the integral extension width.

11 The generic Zagier families

The finite sporadic atlas is best understood against the two infinite families appearing in Zagier's analysis.

11.1 Hypergeometric family

For $d \geq -1/2$ define

$$HG_d : (A, B, \lambda) = (-1, 0, d(d+1)).$$

The distinguished generating function is

$$H_d(z) = {}_2F_1(-d, d+1; 1; -z) = P_d(1+2z),$$

and the recurrence reduces to

$$(n+1)^2 A_{n+1} = (d(d+1) - n(n+1)) A_n.$$

If $d(d+1) \neq 0$, the normalized second solution satisfies

$$B_n = \frac{A_n}{d(d+1)} \quad (n \geq 1). \quad (9)$$

Thus its companion ratio is rationally degenerate. For integral $d \geq 0$ the series terminates.

11.2 Legendrian family and Pfaff equivalence

Define

$$Leg_d : (A, B, \lambda) = (2, 1, d(d+1) + 1).$$

Its first solution is

$$F_d(t) = \frac{{}_2F_1(-d, -d; 1; t)}{(1-t)^{d+1}} = \frac{1}{1-t} P_d\left(\frac{1+t}{1-t}\right).$$

Pfaff's transformation gives the exact identity

$$F_d(t) = \frac{1}{1-t} H_d\left(\frac{t}{1-t}\right). \quad (10)$$

So HG_d and Leg_d are not two different homogeneous objects. They are the same Gauss local system in two Apéry gauges.

Theorem 11.1 (Legendrian companion). *Let A_n be the principal Leg_d solution and $B_0 = 0, B_1 = 1$ the normalized second solution. Then for every $n \geq 1$*

$$A_{n-1}B_n - A_nB_{n-1} = \frac{1}{n^2},$$

so

$$B_n = A_n \sum_{k=1}^n \frac{1}{k^2 A_k A_{k-1}}. \quad (11)$$

Moreover

$$\frac{B_n}{A_n} \rightarrow \xi_d = 2 \sum_{j=0}^{\infty} \frac{(-1)^j}{(j+d+1)^2} = \frac{1}{2} \left[\psi_1\left(\frac{d+1}{2}\right) - \psi_1\left(\frac{d+2}{2}\right) \right]. \quad (12)$$

The parameter recursion is

$$\boxed{\xi_d + \xi_{d+1} = \frac{2}{(d+1)^2}}.$$

Proof. The recurrence is self-adjoint after multiplication by n^2 . The discrete Wronskian of two solutions therefore satisfies a first-order relation whose normalization at $n = 1$ gives $A_{n-1}B_n - A_nB_{n-1} = n^{-2}$. Division by A_nA_{n-1} and telescoping yields (11). The connection value follows from the standard Legendre Q_d connection integral, equivalently by evaluating the resulting alternating Hurwitz series. Separating even and odd terms gives the trigamma expression. Shifting $d \mapsto d + 1$ in the alternating series proves the displayed recursion. $\square$

If $d + 1 = m/q \in \mathbb{Q}_{>0}$, finite Fourier decomposition on residues modulo $2q$ expresses ξ_d as a cyclotomic dilogarithmic period, hence as a finite cyclotomic-linear combination of Dirichlet $L(2, \chi)$ -values. For nonnegative integral m ,

$$\xi_m = (-1)^m \left[\zeta(2) - 2 \sum_{k=1}^m \frac{(-1)^{k-1}}{k^2} \right].$$

In particular $\xi_0 = \zeta(2)$, while the boundary value $d = -1/2$ gives $8G$.

11.3 Why the generic family is not Apéry-exponential

For $d > -1/2$ one has

$$A_n \sim C_d n^d, \quad C_d = \frac{\Gamma(2d+1)}{\Gamma(d+1)^3},$$

and

$$\xi_d - \frac{B_n}{A_n} \sim \frac{1}{C_d^2(2d+1)} n^{-2d-1},$$

so

$$B_n - \xi_d A_n \sim -\frac{1}{C_d(2d+1)} n^{-d-1}.$$

There is no exponential root separation. At $d = -1/2$ the familiar zero-balanced logarithmic resonance appears. Therefore these families do not support the ordinary scalar exponential Apéry mechanism, although this says nothing against vector, Hadamard, determinant or arithmetic-holonomy constructions.

The arithmetic mirror-map subfamily of the zero-balanced Gauss case reduces to the classical rational parameters $a = 1/2, 1/3, 1/4, 1/6$; this is part of the established hypergeometric mirror-map integrality classification [12].

12 Archimedean fold geometry

The source alone does not determine whether B_n/A_n has a real limit. That is a Layer-C question about the location and multiplicity of dominant singularities.

For the twelve holomorphic-source sporadics the canonical realizations split as follows:

$$\begin{aligned} \text{real dominant fold: } & A, C, D, E, F, \alpha, \gamma, \epsilon, \zeta, \\ \text{complex-conjugate dominant folds: } & B, \delta, \eta. \end{aligned}$$

Thus B, δ, η are not “less modular” than the others. Their companions are just as explicit, but coefficient asymptotics inherit oscillation from two equal-modulus complex singularities.

The phenomenon is already visible in Zagier’s row B . The canonical source contains $\frac{1}{2}L(2, \chi_{-3})$, but the global quotient does not converge. Residue-class limits mix that Dirichlet value with a $\zeta(2)$ contribution coming from the second dominant mode. This is exactly the distinction between an intrinsic extension period and an Apéry-coordinate connection constant.

13 Arithmetic atlas

The primary sequences have a remarkable amount of arithmetic structure independent of the companion story. Binomial/constant-term representations are known for the entire fifteen-family census; see in particular Gorodetsky [9]. Lucas congruences for the sporadic sequences were established systematically by Malik–Straub [8], with later refinements through constant-term methods.

A modern two-term supercongruence summary is especially compact. For primes $p \geq 5$ the standard sporadic sequences satisfy congruences of the form

$$A(mp^r) \equiv A(mp^{r-1}) \pmod{p^{\kappa r}},$$

with

$$\kappa = 2 \quad \text{for } B, C, E, F, s_{18}, \quad \kappa = 3 \quad \text{for all other rows in } \mathcal{S}_{15}.$$

The final remaining cases in this census were closed in work culminating with Alinquant–Osburn [11]. These self-sequence supercongruence exponents must not be confused with the denominator weight of the *companion*.

Table 6: Two independent arithmetic invariants.

rows	primary supercongruence exponent κ	companion denominator weight
A	3	2
B, C, E, F	2	2
D	3	2
$\alpha, \gamma, \delta, \epsilon, \zeta, \eta$	3	3
s_7, s_{10}	3	2 sharp
s_{18}	2	2 sharp

The table is a useful test of the three-layer philosophy. Primary Frobenius congruences belong largely to the pure motive/local system, whereas companion denominators measure the integral extension class. There is no reason for the two columns to coincide.

14 Fifteen row-by-row entries

This section records the minimal reference data needed to identify every row without consulting the master tables.

A

Recurrence data $(7, -8, 2)$; level 6; weight-one period; character χ_{-3} at the source. The reverse source is

$$c_A(m) = \sigma_{\text{out}}^{\chi_{-3}, 2}(m) - \sigma_{\text{out}}^{\chi_{-3}, 2}(m/2),$$

with source period $\zeta(2)/4$. The dominant fold is real. Companion denominator weight is 2.

B

Data $(9, 27, 3)$; level 36; χ_{-3} source. The source is

$$c_B(m) = \sigma_{\text{in}}^{\chi_{-3}, 2}(m) - 6\sigma_{\text{in}}^{\chi_{-3}, 2}(m/2) - 8\sigma_{\text{in}}^{\chi_{-3}, 2}(m/4),$$

with intrinsic value $\frac{1}{2}L(2, \chi_{-3})$. The dominant singularities are a complex pair, so no single real coefficient-ratio limit exists. Companion weight 2.

C

Data (10, 9, 3); level 6. The source

$$c_C(m) = \sigma_{\text{in}}^{\chi_{-3}, 2}(m) - 8\sigma_{\text{in}}^{\chi_{-3}, 2}(m/2)$$

has value $\frac{1}{2}L(2, \chi_{-3})$. A useful Lambert form is

$$D^{-2}\Phi_C(q) = \sum_{m \geq 1} \frac{\chi_{-3}(m)}{m^2} \frac{q^m}{1 + q^m}.$$

Companion weight 2.

D

Data (11, -1, 3); the classical Apéry $\zeta(2)$ row. It lives naturally on $\Gamma_1(5)$ rather than one of the eta-quotient $\Gamma_0(N)$ presentations. The coordinate is (6); the source is the quartic-character weight-three Eisenstein combination in Table 3; source/connection value $\zeta(2)/5$; companion weight 2.

E

Data (12, 32, 4); level 8; source character χ_{-4} . Its critical source value is

$$\frac{1}{2}L(2, \chi_{-4}) = \frac{G}{2},$$

where G is Catalan's constant. The row is therefore a canonical control case for any extension of Apéry methods to Catalan's constant. Nothing in the source theorem alone proves irrationality of G . Companion weight 2.

F

Data (17, 72, 6); level 12; χ_{-3} source; critical value $\frac{5}{8}L(2, \chi_{-3})$. Companion weight 2.

α

Cubic data (10, 4, 64); partner of C ; level 12; untwisted old- E_4 source. It is a full period projector and has source value $7\zeta(3)/24$. Companion weight 3.

γ

Cubic data (17, 5, 1); partner of F ; level 6. This is Apéry's $\zeta(3)$ recurrence. Its source, Dirichlet multiplier and explicit trilogarithmic companion are given in Section 7. Companion weight 3.

δ

Cubic data (7, 3, 81); partner of A ; level 12. Its source value is $13\zeta(3)/54$, but its oldform polynomial retains nonzero $P(2)$ and $P(4)$ and its canonical dominant folds are complex. It is therefore the cleanest example showing that “target $\zeta(3)$ ”, “full period projector”, and “ordinary Apéry ratio” are three different properties.

ϵ

Cubic data (12, 4, 16); partner of E ; level 8; full untwisted projector; source value $7\zeta(3)/32$; companion weight 3.

ζ

Cubic data $(9, 3, -27)$; partner of B ; level 9. The source belongs to the $\chi_{-3} \otimes \chi_{-3}$ Eisenstein sector and has critical value

$$\frac{1}{3}L(3, \chi_{-3}) = \frac{4\pi^3}{243\sqrt{3}}.$$

It is character-twisted even though the total Nebentypus is trivial.

η

Cubic data $(11, 5, 125)$; partner of D ; level 20. The χ_5 -twisted source has critical value $\frac{1}{2}L(3, \chi_5)$. Its canonical dominant folds are complex-conjugate; companion weight 3.

s_7

Data $(13, 4, -27, 3)$ in convention (2); level 7. Cooper's rational-Hauptmodul parametrization places the dominant singularity $t = 1/27$ at the Fricke fixed point. The source is exact but belongs to the generalized Cooper stratum. Companion weight 2 is sharp. The connection $\zeta(2)/7$ is recorded as numerical/conjectural here.

s_{10}

Data $(6, 2, -64, 4)$; level 10; fourth Franel sequence $\sum_k \binom{n}{k}^4$. Companion weight 2 is sharp. Its Apéry limit $\zeta(2)/5$ follows from the general Franel-power analysis of [10].

s_{18}

Data $(14, 6, 192, -12)$; level 18; generalized Cooper source; sharp companion weight 2. High-precision computation identifies the connection constant with $\frac{1}{2}L(2, \chi_{-3})$, but an exact regularized Eichler proof is still required.

15 A three-layer periodic table

The atlas can now be compressed conceptually.

Table 7: Three structural classes of the fifteen-family census.

class		rows	interpretation
Holomorphic Eisenstein + real fold	Eisen-	$A, C, D, E, F, \alpha, \gamma, \epsilon, \zeta$	Exact pure Eisenstein extension and an ordinary real connection geometry.
Holomorphic Eisenstein + complex fold	Eisen-	B, δ, η	Same exact extension theory, but coefficient asymptotics oscillate because two dominant folds have equal modulus.
Generalized symmetric square	Cooper	s_7, s_{10}, s_{18}	Exact $\Phi = FDt$, rational-Hauptmodul geometry, possible meromorphic source, and one arithmetically free integration; sharp companion weight 2.

At Layer A, six of the first twelve rows pair into six rank-two geometries. At Layer B, old-form/Eisenstein data distinguish the normal functions and their critical periods. At Layer C, the choice of coordinate and singularity geometry decides integrality, denominator cost and whether a simple coefficient-ratio limit exists.

This is the sense in which the sporadic census becomes understandable rather than merely memorable.

16 Why “sporadic” is a structural phenomenon

The generic families show what is missing away from the isolated modular points. The HG_d and Leg_d families already possess a perfectly respectable rank-two Gauss local system, and for rational parameters their connection constants can even be cyclotomic L -values. But their Layer-C behavior is degenerate: either the normalized companion is rationally proportional to the principal solution, or the two asymptotic solutions have the same exponential modulus and differ only polynomially.

The sporadic rows instead align three constraints simultaneously:

- (1) a rigid modular/elliptic homogeneous local system;
- (2) a nontrivial Eisenstein or regularized modular extension carrying a critical L -value;
- (3) an Apéry coordinate whose singularity geometry can create useful analytic separation without destroying integrality.

No single one of these conditions is exceptional. Their simultaneous occurrence is.

This perspective also explains why an irrationality proof is a strictly stronger event than the identification of a source. The modular source controls the period content; the integral extension controls denominator width; and the singularity geometry controls archimedean decay. A proof succeeds only when all three inequalities align.

17 The converse census problem

The present paper proves a structural theorem for the standard known census. A stronger result would make the reverse machine enumerative.

Conjecture 17.1 (Reverse-census principle). *Fix a normalized Apéry recurrence class and impose:*

- (i) *an integral principal solution and integral canonical mirror map after a fixed rational scaling;*
- (ii) *a genus-zero or finite modular realization of the homogeneous local system;*
- (iii) *an Eisenstein/polylogarithmic extension of bounded weight whose oldform projector has the required boundary vanishing;*
- (iv) *finite arithmetic extension width.*

Then the isolated scalar realizations form a finite set obtained from the known modular source classes by algebraic pullback, isogeny, gauge and extension-coboundary operations, together with explicitly classified hypergeometric degenerations.

This formulation is intentionally more invariant than “solve for integer triples (A, B, λ) ”. It suggests a proof strategy: classify the permitted Layer-A local systems, classify finite Eisenstein projectors at Layer B, and only then solve the integrality constraints of Layer C. The classical fifteen-family atlas is the first large finite test of that program.

18 Outlook: a reusable application protocol

The same procedure can be applied to other identity corpora, including Ramanujan-Machine continued fractions and modular special-value recurrences:

recurrence/series $\rightarrow (t, F) \rightarrow \Phi = F D t \rightarrow$ Eisenstein/Hecke decomposition $\rightarrow$ critical period $\rightarrow$ extension and ar

The value of the sporadic census is therefore methodological as well as historical: it is a controlled benchmark in which every layer can be seen explicitly and compared across many examples.

A Finite certification of the eleven eta rows

For each eta quotient $\prod_{d|N} \eta(d\tau)^{r_d}$, the standard eta transformation law and Ligozat cusp-order formula certify weight, character and holomorphy. The Wronskian identity then fixes the normalized principal period, so FDt is a modular form of the predicted weight and character. The candidate Eisenstein series in Tables 3–4 belongs to the same finite-dimensional space. Equality follows by a finite Sturm comparison.

A conservative set of Sturm bounds for the eleven eta rows is

row	A	B	C	E	F	α	γ	δ	ϵ	ζ	η
bound	3	18	3	3	6	8	4	8	4	4	12.

The D row is handled separately on $\Gamma_1(5)$ with the quartic-character weight-three Eisenstein space. Thus no identity in the twelve-row source theorem is being inferred from an indefinite numerical q -expansion.

B Polylogarithmic companions for the untwisted cubic rows

Put

$$\mathcal{L}_3(q) = \sum_{m \geq 1} \frac{\sigma_3(m)}{m^3} q^m.$$

Then direct oldform integration gives

$$\begin{aligned} \Theta_\alpha &= \mathcal{L}_3(q) - \frac{17}{8}\mathcal{L}_3(q^2) - \frac{1}{3}\mathcal{L}_3(q^3) + \frac{1}{4}\mathcal{L}_3(q^4) + \frac{17}{24}\mathcal{L}_3(q^6) - \frac{1}{12}\mathcal{L}_3(q^{12}), \\ \Theta_\gamma &= \mathcal{L}_3(q) - \frac{7}{2}\mathcal{L}_3(q^2) + \frac{7}{3}\mathcal{L}_3(q^3) - \frac{1}{6}\mathcal{L}_3(q^6), \\ \Theta_\delta &= \mathcal{L}_3(q) - \frac{7}{4}\mathcal{L}_3(q^2) - \frac{1}{27}\mathcal{L}_3(q^3) + \frac{1}{4}\mathcal{L}_3(q^4) + \frac{7}{108}\mathcal{L}_3(q^6) - \frac{1}{108}\mathcal{L}_3(q^{12}), \\ \Theta_\epsilon &= \mathcal{L}_3(q) - \frac{21}{8}\mathcal{L}_3(q^2) + \frac{21}{16}\mathcal{L}_3(q^4) - \frac{1}{8}\mathcal{L}_3(q^8). \end{aligned}$$

The actual companion is $F\Theta_f$. Character-twisted analogues for ζ and η are obtained by replacing σ_3 with the corresponding twisted divisor sums.

C Evidence ledger

For publication use, the logical status of the main ingredients is:

statement	status
Twelve source identities $\Phi = FDt$ and pure Eisenstein decompositions	exact modular theorem; finite eta/Ligozat/Wronskian/Sturm certificates.
Six pairings and classical cubic symmetric-square identities	exact algebraic identities.
General Cooper symmetric-square identity	exact algebraic identity in a, b, c, d .
Cooper free first primitive and sharp d_n^2 weight	exact theorem in the project ledger; sharpness visible at $n = 2$.
s_{10} connection $\zeta(2)/5$	published, via Franel-power Apéry limits.
s_7 connection $\zeta(2)/7$	numerical/connection conjecture in this atlas.
s_{18} connection $\frac{1}{2}L(2, \chi_{-3})$	numerical/connection conjecture in this atlas.
HG_d/Leg_d Pfaff equivalence, Casoratian, trigamma limit, asymptotics	exact analytic identities.
Global exhaustion of all integral recurrences	not claimed; converse census remains open.

D A compact machine-readable dictionary

For convenient reuse, the canonical twelve-row source laws are determined by the tuples and divisor sums in Tables 3–4, while the recurrence pairing is determined by (4). These are sufficient to regenerate the critical-value table because oldform dilation acts by d^{-s} on Dirichlet series. The accompanying source package contains a short exact-arithmetic script checking the cubic oldform fingerprints, the six pairings, and the Cooper sharpness values.

References

- [1] Apéry Systems Project, *A Three-Layer Classification of Dirichlet L-Value Realizations: Hecke–Eisenstein Sources, Extension Geometry, and Apéry Realizations*, manuscript, 2026.
- [2] R. Apéry, *Irrationalité de $\zeta(2)$ et $\zeta(3)$* , *Astérisque* **61** (1979), 11–13.
- [3] F. Beukers, *Irrationality proofs using modular forms*, *Astérisque* **147–148** (1987), 271–283.
- [4] D. Zagier, *Integral solutions of Apéry-like recurrence equations*, in *Groups and Symmetries: From Neolithic Scots to John McKay*, CRM Proc. Lecture Notes **47**, Amer. Math. Soc., 2009, 349–366.
- [5] Y. Yang, *Apéry limits and special values of L-functions*, *J. Math. Anal. Appl.* **343** (2008), 492–513.
- [6] G. Almkvist and W. Zudilin, *Differential equations, mirror maps and zeta values*, in *Mirror Symmetry V*, AMS/IP Stud. Adv. Math. **38**, Amer. Math. Soc., 2006, 481–515.
- [7] S. Cooper, *Sporadic sequences, modular forms and new series for $1/\pi$* , *Ramanujan J.* **29** (2012), 163–183.
- [8] A. Malik and A. Straub, *Divisibility properties of sporadic Apéry-like numbers*, *Res. Number Theory* **2** (2016), Art. 5.
- [9] O. Gorodetsky, *New representations for all sporadic Apéry-like sequences, with applications to congruences*, *Experiment. Math.* **32** (2023), 641–656.
- [10] A. Straub and W. Zudilin, *Sums of powers of binomials, their Apéry limits, and Franel’s suspicions*, *Int. Math. Res. Not. IMRN* **2023** (2023), no. 11, 9861–9879.
- [11] B. Alinquant and R. Osburn, *On sporadic sequences*, *Monatsh. Math.* **205** (2024), 217–234.
- [12] H. Movasati and K. M. Shokri, *Modular-type functions attached to Calabi–Yau varieties: integrality properties*, preprint arXiv:1306.5662 (2013).